

A Phase-Free Critical–First-Lower Actual Signed-Completion Gap

Bin Wang

August 2026

Abstract

RH-352 proves actual natural-scale cancellation on the growing lower-even ladder but excludes the critical order $2k$ and first lower sideband $2k - 2$. We close those two normalized signed-completion coordinates on the same physical clock. Put $m = k - 1$, $H_\ell = \ell R^{-2\ell}$, and $x = (\beta R)^2$. The source-locked direct coefficients are

$$p_k^0 = Y_k^0 + \mathcal{P}_k^0 - S_k^0, \quad p_k^- = Y_k^- + \mathcal{P}_k^- - S_k^-.$$

The actual modulus-complement cap and deterministic all-order envelope give

$$\limsup_{k \rightarrow \infty} \left(\max \left\{ \frac{|p_k^0|}{2H_k x^k}, \frac{|p_k^-|}{2H_m x^m} \right\} \right)^{1/k} \leq \max \left\{ \frac{r_H^2 \lambda^3}{4}, \frac{1}{\lambda} \right\} < 1.$$

Writing $\gamma_k = C_* C_M \lambda^{\eta_k}$, the critical and first-lower source laws then force the actual normalized remainders

$$Z_k^0 := \frac{C_M Y_k^0}{2H_k x^k} = 2 - \gamma_k + o(1), \quad Z_k^- := \frac{C_M Y_k^-}{2H_m x^m} = 1 - \frac{\gamma_k}{\lambda} + o(1).$$

The physical phase cancels exactly across the two orders:

$$Z_k^0 - \lambda Z_k^- \longrightarrow 2 - \lambda > \frac{3}{10}.$$

Consequently

$$\liminf_{k \rightarrow \infty} \max\{|Z_k^0|, |Z_k^-|\} \geq \frac{2 - \lambda}{1 + \lambda} > \frac{1}{9}.$$

Thus the two-coordinate maximum of the actual boundary remainders supplies an exponentially large unnormalized weighted contribution; the maximizing order may depend on k . This is signed cancellation supply, not direct-prefix divergence: $p_k^0 = o(H_k)$ and $p_k^- = o(H_m)$ remain open. Odd and upper-alias orders, full E_{off} , RH-288, Gates A–E, and all Riemann-hypothesis claims remain open.

1 One clock and two excluded boundary orders

Use the physical natural clock and bounded phase

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_k = k - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad \sup_k |\eta_k| < \infty. \quad (1)$$

Fix

$$r_H = \frac{17}{20}, \quad q = \frac{1}{2}, \quad R = \frac{7}{5}, \quad \frac{28}{17} < \lambda < \frac{17}{10}, \quad (2)$$

and put

$$\beta = \frac{1}{r_H \sqrt{\lambda}}, \quad q_* = \frac{1}{r_H \lambda}, \quad x = (\beta R)^2 > 1, \quad H_\ell = \ell R^{-2\ell}. \quad (3)$$

The strict multiplier bracket is source-certified in RH-262 and RH-336 [1, 8].

Set

$$m = k - 1. \quad (4)$$

This is one noise sequence. The symbol m is only the period parameter of the first-lower boundary orbit.

At the critical order, RH-344 and RH-345 give the exact direct coefficient

$$p_k^0 := p_{\sigma,k,2k} = Y_k^0 + \mathcal{P}_k^0 - S_k^0, \quad Y_k^0 = \mathcal{T}_k^{\text{rest}} - d_{\sigma,k,2k}, \quad S_k^0 = \mathcal{A}_{k,2k} + F_k^{\text{orb}}. \quad (5)$$

Their deterministic and parity laws are

$$S_k^0 = \frac{4k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (6)$$

$$\frac{\mathcal{P}_k^0}{\mathcal{A}_{k,2k}} = C_* C_M \lambda^{\eta_k} \{1 + o(1)\}, \quad \mathcal{A}_{k,2k} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}. \quad (7)$$

At the first lower sideband, RH-346 and RH-347 give

$$p_k^- := p_{\sigma,k,2m} = Y_k^- + \mathcal{P}_k^- - S_k^-, \quad Y_k^- = \mathcal{T}_{k,m}^{\text{rest}} - d_{\sigma,k,2m}, \quad S_k^- = F_m^{\text{orb}} + \mathcal{A}_{k,2m}, \quad (8)$$

with

$$S_k^- = \frac{2m}{C_M} \beta^{2m} \{1 + o(1)\}, \quad (9)$$

$$\frac{\mathcal{P}_k^-}{S_k^-} = C_* C_M \lambda^{\eta_k-1} \{1 + o(1)\}. \quad (10)$$

All four quantities $Y_k^0, Y_k^-, p_k^0, p_k^-$ are actual signed coefficients or remainders in one physical data type. They are not scalar completions or synthetic operators [2–4, 6].

2 The actual two-order direct cap

The direct coefficient has the source-locked representation

$$p_{\sigma,k,n} = \tau_{\sigma,n} - a_n. \quad (11)$$

RH-282 and RH-267 prove, for all $n \geq 2$,

$$|\tau_{\sigma,n}| \leq \sigma^{-1} q^{n-2}, \quad |a_n| < 48 q_*^n. \quad (12)$$

These are actual modulus-complement and deterministic-target bounds, respectively; RH-288 and RH-340 identify their difference as the direct coefficient in (11) [5, 7, 9–11].

Define

$$V_k = \max \left\{ \frac{|p_k^0|}{2H_k x^k}, \frac{|p_k^-|}{2H_m x^m} \right\}, \quad \rho_N = \frac{r_H^2 \lambda^3}{4}, \quad \rho_T = \frac{1}{\lambda}. \quad (13)$$

Theorem 2.1 (Actual two-order natural-scale cap). *One has*

$$\boxed{\limsup_{k \rightarrow \infty} V_k^{1/k} \leq \max\{\rho_N, \rho_T\} < 1.} \quad (14)$$

More precisely,

$$\rho_N < \frac{1419857}{1600000} < 1, \quad \rho_T < \frac{17}{28} < 1. \quad (15)$$

Proof. For $\ell = k$ or $\ell = m = k - 1$, the identities $2H_\ell x^\ell = 2\ell\beta^{2\ell}$ and $\sigma^{-1} = \lambda^{2(k-\eta_k)}$ give

$$\frac{|\tau_{\sigma, 2\ell}|}{2H_\ell x^\ell} \leq \frac{2}{\ell} \sigma^{-1} (q^2 r_H^2 \lambda)^\ell, \quad (16)$$

$$\frac{|a_{2\ell}|}{2H_\ell x^\ell} < \frac{24}{\ell} \lambda^{-\ell}. \quad (17)$$

For $\ell = k$, their k th-root ceilings are ρ_N and ρ_T . Replacing k by $k - 1$ changes only fixed factors and has the same root ceilings. The triangle inequality in (11) proves (14). The strict rational bounds follow from (2), exactly as in RH-352. $\square$

The theorem is normalized at the leading boundary-demand scale. It does not imply $p_k^0 = o(H_k)$ or $p_k^- = o(H_m)$ after the factors x^k and x^m are removed.

3 Actual completion laws and phase elimination

Put

$$\gamma_k = C_* C_M \lambda^{\eta_k}. \quad (18)$$

Bounded phase makes γ_k bounded above and away from zero. Define the actual normalized signed remainders

$$Z_k^0 = \frac{C_M Y_k^0}{2H_k x^k}, \quad Z_k^- = \frac{C_M Y_k^-}{2H_m x^m}. \quad (19)$$

Theorem 3.1 (Actual critical and first-lower completion laws). *On every bounded-phase physical clock,*

$$\boxed{Z_k^0 = 2 - \gamma_k + o(1),} \quad (20)$$

$$\boxed{Z_k^- = 1 - \frac{\gamma_k}{\lambda} + o(1).} \quad (21)$$

Proof. Since $2H_k x^k = 2k\beta^{2k}$, (6) gives

$$\frac{C_M S_k^0}{2H_k x^k} \longrightarrow 2.$$

Equations (7) and boundedness of γ_k give

$$\frac{C_M \mathcal{P}_k^0}{2H_k x^k} = \gamma_k + o(1).$$

Solve (5) for Y_k^0 and use theorem 2.1 to obtain (20).

Likewise $2H_m x^m = 2m\beta^{2m}$, so (9)–(10) imply

$$\frac{C_M S_k^-}{2H_m x^m} \longrightarrow 1, \quad \frac{C_M \mathcal{P}_k^-}{2H_m x^m} = \frac{\gamma_k}{\lambda} + o(1).$$

Solving (8) and applying theorem 2.1 proves (21). $\square$

The two leading completion laws contain the same physical phase scalar. Their difference eliminates it.

Theorem 3.2 (Phase-free actual boundary gap). *The actual remainder pair satisfies*

$$\boxed{Z_k^0 - \lambda Z_k^- \longrightarrow 2 - \lambda > \frac{3}{10}.} \quad (22)$$

Consequently

$$\liminf_{k \rightarrow \infty} \max\{|Z_k^0|, |Z_k^-|\} \geq \frac{2-\lambda}{1+\lambda} > \frac{1}{9}. \quad (23)$$

The constant is sharp for the leading affine family: equality in the minimax problem is attained at

$$\gamma_* = \frac{3\lambda}{1+\lambda}, \quad 2 - \gamma_* = \frac{2-\lambda}{1+\lambda}, \quad 1 - \frac{\gamma_*}{\lambda} = -\frac{2-\lambda}{1+\lambda}. \quad (24)$$

Proof. Subtract λ times (21) from (20); the two occurrences of γ_k cancel. The upper multiplier bound gives $2 - \lambda > 3/10$.

For arbitrary real u, v ,

$$|u - \lambda v| \leq |u| + \lambda|v| \leq (1 + \lambda) \max\{|u|, |v|\}.$$

Apply this to $u = Z_k^0$, $v = Z_k^-$ and pass to the lower limit. Since $\lambda < 17/10$ and $t \mapsto (2-t)/(1+t)$ is decreasing,

$$\frac{2-\lambda}{1+\lambda} > \frac{2-17/10}{1+17/10} = \frac{1}{9}.$$

Direct substitution verifies (24), so the leading minimax constant is sharp. $\square$

Corollary 3.3 (Forced actual boundary signed supply). *The maximum of the two actual remainder contributions is exponentially large; the maximizing coordinate may vary with k . More precisely,*

$$\liminf_{k \rightarrow \infty} \frac{C_M}{x^{k-1}} \max\left\{\frac{|Y_k^0|}{2H_k}, \frac{|Y_k^-|}{2H_m}\right\} \geq \frac{2-\lambda}{1+\lambda} > \frac{1}{9}. \quad (25)$$

Proof. The two unnormalized quantities are respectively $x^k|Z_k^0|/C_M$ and $x^{k-1}|Z_k^-|/C_M$. Since $x > 1$, their maximum is at least $x^{k-1} \max\{|Z_k^0|, |Z_k^-|\}/C_M$. Apply (23). $\square$

This divergence belongs to the actual signed remainders Y , not to the direct coefficients p . It is the leading supply that cancels the deterministic/parity packets in (5) and (8).

4 Executable protocol and claim boundary

The artifact checks the exact affine identity $(2-\gamma) - \lambda(1-\gamma/\lambda) = 2-\lambda$, the exact minimax optimizer, the global rational bounds $2-\lambda > 3/10$ and $(2-\lambda)/(1+\lambda) > 1/9$, and the two natural-scale rate certificates. Its finite rows use the in-range rational fixture $\lambda = 5/3$. They reproduce formulas only; they are not observations of p , Y , or the noisy operator and are not asymptotic evidence.

RH-353 proves actual leading natural-scale completion at the critical and first-lower orders and a phase-free cross-order remainder gap. It does not prove $p_k^0 = o(H_k)$ or $p_k^- = o(H_m)$, control odd or upper-alias orders, decide the full E_{off} aggregate, prove head transport, or close the RH-241 moving noisy all-order envelope and coefficient bridge. RH-288 is inactive. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt-weighted prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

- [1] Bin Wang. A certified deterministic-numerator boundary budget, 2026. Repository paper RH-262.
- [2] Bin Wang. Complete critical boundary-orbit atom decomposition, 2026. Repository paper RH-344.

- [3] Bin Wang. Complete lower-sideband boundary-orbit decomposition, 2026. Repository paper RH-346.
- [4] Bin Wang. Double-alias parity-phase compensation obstruction, 2026. Repository paper RH-345.
- [5] Bin Wang. A weighted prefix–tail determinant gluing criterion, 2026. Repository paper RH-288.
- [6] Bin Wang. Lower-sideband scalar balance underdetermination, 2026. Repository paper RH-347.
- [7] Bin Wang. Modulus-cap-forced growing-ladder signed cancellation, 2026. Repository paper RH-352.
- [8] Bin Wang. Projector-mass first-alias threshold and an isospectral cell obstruction, 2026. Repository paper RH-336.
- [9] Bin Wang. A modulus-complete spectral-tail certificate, 2026. Repository paper RH-282.
- [10] Bin Wang. Synchronized determinant prefixes and a two-order orbit–head compensation obstruction, 2026. Repository paper RH-340.
- [11] Bin Wang. A certified unified deterministic trace envelope, 2026. Repository paper RH-267.